

Phase 3: Project Design Deliverables

Document 02: Proposed Solution Workflow & Technical Automation Specification

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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1. Solution Overview & Architectural Principles

The **Automated Standard Laptop Procurement Solution** replaces disparate, manual email interactions with an enterprise-grade, event-driven orchestration architecture built natively within **ServiceNow Flow Designer**. The solution establishes a standardized digital bridge between employee self-service in the Service Portal / Employee Center, management approval governance, operational hardware staging by IT fulfillment teams, physical deskside/ remote delivery logistics, and automated Hardware Asset Management (**alm_hardware**) lifecycle synchronization.

```
flowchart TD
    subgraph Trigger_Phase [1. Ingestion & Trigger Phase]
        T1[Service Portal Catalog Submission] --> T2[Insert sc_request & sc_req_item]
        T2 --> T3[Flow Designer Trigger Activated]
    end

    subgraph Governance_Phase [2. Approval & Governance Phase]
        T3 --> G1[Validate Manager Hierarchy]
        G1 --> G2[Ask for Approval Action]
        G2 --> G3{Manager Decision?}
        G3 -->|Approved| F1[Stage: Fulfillment]
        G3 -->|Rejected| R1[Stage: Request Cancelled]
    end

    subgraph Fulfillment_Phase [3. Automated Multi-Stage Fulfillment]
        F1 --> F2[Create Task 1: Stage & Image Hardware]
        F2 --> F3[Wait for Task 1 Closure & Barcode Scan]
        F3 --> F4[Create Task 2: Ship & Deskside Logistics]
        F4 --> F5[Wait for Task 2 Closure & Delivery Proof]
    end

    subgraph Closure_Phase [4. Asset Sync & Lifecycle Closure]
        F5 --> C1[Update alm_hardware: install_status = In Use]
```

```

C1 --> C2[Link cmdb_ci_computer to RITM]
C2 --> C3[Close sc_req_item & sc_request]
C3 --> C4[Dispatch Delivery & Welcome Email]
R1 --> C5[Close RITM/REQ Incomplete & Send Rejection Notice]
end

subgraph Error_Catch [5. Global Flow Error Handler]
  ERR[Runtime Exception / DB Timeout] --> E1[Log Context Syslog]
  E1 --> E2[Create P2 Incident for Platform Support]
  E2 --> E3[Annotate RITM Work Notes]
end

Trigger_Phase -->|Exception| Error_Catch
Governance_Phase -->|Exception| Error_Catch
Fulfillment_Phase -->|Exception| Error_Catch
Closure_Phase -->|Exception| Error_Catch

```

Core Architecture & Engineering Principles

- 1. Zero Core Modifications & 100% OOTB Upgradability:** Utilizes standard ITSM schema (`sc_request`, `sc_req_item`, `sc_task`, `sysapproval_approver`, `alm_hardware`). Zero legacy custom tables or deprecated workflow engine dependencies (`wf_workflow`).
- 2. Elevated Engine Security Execution:** Flow execution runs in the background as `System User`, ensuring that cross-table transactional updates (e.g., updating asset status and provisioning fulfillment tasks) succeed reliably regardless of the restricted end-user ACLs of the catalog requester.
- 3. Defensive Governance & Anti-Hang Routing:** Pre-decision validation blocks verify the presence and active status of line managers before routing, routing to a fallback governance queue if null or terminated.
- 4. Two-Stage Sequential Task Decomposition:** Isolates technical imaging from physical delivery logistics, preventing premature ticket closure and lost computing devices.
- 5. Native Fault Tolerance:** Encapsulated within a dedicated Flow Designer Error Handler block that logs execution IDs, alerts platform administrators, and generates high-priority incidents automatically upon unhandled exceptions.

2. Trigger Mechanics & Context Instantiation

The execution lifecycle begins synchronously upon database insertion of the line-item record and transitions immediately to asynchronous background processing.

2.1 Trigger Definition Specification

Parameter	Configuration Value	Architectural Rationale
Trigger Type	<code>Record Created</code>	Captures newly submitted catalog line items immediately upon transaction commit.
Table	<code>sc_req_item</code> (Requested Item)	Scopes execution directly to the individual item context rather than the generic cart header (<code>sc_request</code>).

Conditions (Filter)	<code>cat_item IS Standard Laptop Order (sys_id: alb2c3d4e5f60718293a4b5c6d7e8f90)
AND request.request_state IS NOT Closed Cancelled</code>	Restricts flow invocation strictly to the standardized laptop offering, preventing trigger collisions with software or accessory requests.
Run When	<code>Record created in database</code>	Ensures database row integrity before spawning flow worker threads.
Execution Environment	<code>Background (Asynchronous)</code>	Prevents user UI thread latency; executes on ServiceNow worker node threads (<code>glide.scheduler.worker</code>).
Run With Roles	<code>System User</code>	Elevates execution context privileges to enable writes across <code>sc_task</code> , <code>alm_hardware</code> , and <code>sys_audit</code> .

2.2 Data Pill Extraction & Context Mapping

Upon instantiation, the Flow Designer engine constructs the `sys_flow_context` object and resolves data pills accessible across all downstream action steps:

```
[Trigger Object: trigger.current]
├─ sys_id                (GUID: Unique identifier of sc_req_item)
├─ number                (String: e.g., "RITM0010501")
├─ request                (Reference -> sc_request: Parent Cart Header)
│   └─ number            (String: e.g., "REQ0010201")
│       └─ requested_for (Reference -> sys_user: Target Beneficiary)
├─ opened_by             (Reference -> sys_user: Submitting User)
│   └─ sys_id            (GUID)
│       └─ name           (String: e.g., "Jane Smith")
│           └─ email       (String: e.g., "jane.smith@enterprise.com")
│               └─ manager (Reference -> sys_user: Direct Line Manager)
│                   └─ sys_id (GUID)
│                       └─ name (String: e.g., "Robert Davis")
│                           └─ email (String: e.g., "robert.davis@enterprise.com")
│                               └─ active (Boolean: True / False)
│                                   └─ department (Reference -> cmn_department)
│                                       └─ dept_head (Reference -> sys_user: Department Head)
└─ variables              (Catalog Item Variable Set)
    └─ laptop_model        (Choice: Developer 16" / Business 14" / Ultralight 13")
        └─ operating_system (Choice: Windows 11 Enterprise / macOS Sonoma / Ubuntu LTS)
            └─ ram_storage_bundle (Choice: Standard 16GB/512GB / Performance 32GB/1TB)
                └─ accessories (List / Multi-Choice: USB-C Dock, Dual 27" Monitors, Backpack)
                    └─ delivery_type (Choice: Deskside Handover / Remote Home Delivery)
                        └─ shipping_address (Multi-line String: Verified Residential / Office Address)
                            └─ contact_phone (String: E.164 Formatted Contact Number)
                                └─ business_justification (Multi-line String: Business Justification Narrative)
```

3. End-to-End Automated Workflow Specification (17 Discrete Operational

Steps)

The end-to-end execution path consists of 17 discrete, sequentially orchestrated operational steps from initial trigger instantiation through final lifecycle decommissioning.

```
sequenceDiagram
    autonumber
    actor Requester as Employee / Requester
    participant Flow as Flow Designer Engine
    participant RITM as sc_req_item
    participant REQ as sc_request
    actor Approver as Line Manager
    participant Task1 as sc_task (Staging)
    participant Task2 as sc_task (Logistics)
    participant Asset as alm_hardware
    actor Tech as Hardware Tech
    actor Logistics as Logistics Agent

    Requester->>RITM: Submits Laptop Request
    RITM-->>Flow: Step 1: Fire Trigger (Record Created)
    Flow->>RITM: Step 2: Set Stage = waiting_for_approval, State = 1
    Flow->>Requester: Step 3: Send Submission Receipt Email (NOTIF-01)
    Flow->>Flow: Step 4: Validate Manager Presence (Null Check)
    Flow->>Flow: Step 5: Route Approval Target
    Flow->>Approver: Step 6: Dispatch Approval Request Email (NOTIF-02)
    Flow->>Approver: Step 7: Execute Ask for Approval (sysapproval_approver)
    Flow->>Approver: Step 8: Monitor SLA Timers (T+48h Reminder, T+120h Escalate)
    Flow->>Flow: Step 9: Conditional Decision on Approval State

    alt Path A: Approved
        Approver->>Flow: Approves Request
        Flow->>RITM: Step 10: Set Stage = fulfillment, State = 2 (WIP)
        Flow->>Requester: Step 11: Send Approval Confirmation Email
        Flow->>Task1: Step 12: Create Task 1 (Hardware Staging & Imaging)
        Tech->>Task1: Reserve Asset, Image OS, Scan Serial, Close Complete
        Task1-->>Flow: Step 13: Wait for Condition (Task 1 Complete)
        Flow->>Task2: Step 14: Create Task 2 (Logistics & Delivery Dispatch)
        Logistics->>Task2: Ship Device, Record Tracking #, Confirm Handover
        Task2-->>Flow: Step 15: Wait for Condition (Task 2 Complete)
        Flow->>Asset: Step 16: Update alm_hardware (In Use, Assigned)
        Flow->>RITM: Step 17A: Close RITM (Closed Complete)
        Flow->>REQ: Close Parent REQ (Closed Complete)
        Flow->>Requester: Dispatch Delivery & Welcome Email (NOTIF-05)
    else Path B: Rejected
        Approver->>Flow: Rejects Request with Comments
        Flow->>RITM: Step 17B: Set Stage = request_cancelled, State = 4
        Flow->>REQ: Set REQ = closed_incomplete
        Flow->>Requester: Dispatch Rejection Advisory Email (NOTIF-03)
    end
```

Detailed Operational Step Walkthrough

Step 1: Trigger Evaluation & Flow Instantiation

- **Execution Action:** Engine event listener intercepts insertion of `sc_req_item`.
- **Validation Logic:** Validates that `cat_item.name == 'Standard Laptop Order'` and `request.request_state != 'Closed Complete'`

```
= 'closed_cancelled'.
```

- **Output:** Generates a dedicated `sys_flow_context` record (State: In Progress).

Step 2: Initialize Requested Item Record State

- **Action:** Update Record on `sc_req_item`.
- **Field Modifications:**

```
• stage = waiting_for_approval
• state = 1 (Open)
• approval = requested
• work_notes = Automated Laptop Procurement flow instantiated. Evaluating line manager approval routing.
```

Step 3: Dispatch Submission Acknowledgment Notification (NOTIF-01)

- **Action:** Send Email
- **Recipient:** `trigger.current.requested_for.email`
- **Subject:** Order Confirmation: Standard Laptop Request - {trigger.current.number}
- **Payload:** Formatted HTML containing RITM number, selected laptop model, configuration specifications, target delivery address, and the 72-hour fulfillment commitment timeline.

Step 4: Decision Logic — Validate Line Manager Presence & Account Health

- **Action:** Flow Logic -> Decision / If Branch
- **Condition:**

```
(trigger.current.opened_by.manager != nil) &&
(trigger.current.opened_by.manager.active == true) &&
(trigger.current.opened_by.manager.email != nil)
```

- **Branching Paths:**
- *If True:* Proceed to Step 5 (Standard Routing).
- *If False:* Divert to Governance Exception Handling (Step 5 Fallback).

Step 5: Route Approval Target & Handle Manager Absence

- **Path 5A (Standard):** Target Approver is assigned as `trigger.current.opened_by.manager`.
- **Path 5B (Governance Fallback):**
- Target Approver is dynamically assigned to Assignment Group: IT Procurement Approvers (`sys_user_group`).
- Action Update Record logs high-priority Work Note: SECURITY ALERT: Requester has no active manager in `sys_user`. Order escalated to IT Procurement Governance queue.
- Flag approval as Escalated.

Step 6: Dispatch Manager Approval Request Notification (NOTIF-02)

- **Action:** Send Email
- **Recipient:** Resolved Approver (Manager or Governance Group).
- **Subject:** ACTION REQUIRED: Laptop Procurement Approval Request for {trigger.current.requested_for.name} - {trigger.current.number}
- **Payload:** Actionable HTML template with embedded corporate cost center, business justification, laptop model, total expenditure, and interactive one-click "Approve" / "Reject" response tokens.

Step 7: Execute Ask for Approval Action

- **Action:** Ask for Approval
- **Target Record:** `trigger.current (sc_req_item)`
- **Approval Rules:**
 - Approve when: Anyone approves
 - Reject when: Anyone rejects
- **Due Date Rule:** Dynamic calculation set to 5 Business Days from record creation.

Step 8: SLA Reminder & Dynamic Escalation Subflow

- **Action:** Flow Logic -> Parallel Flow / Timer Execution
- **Timer 1 (T+48 Business Hours):**
 - If approval state remains `requested`, dispatch reminder notification to approver: REMINDER: Laptop Approval Pending for `{trigger.current.number}`.
- **Timer 2 (T+120 Business Hours / 5 Days):**
 - If approval state remains `requested`, execute auto-escalation: reassign approval to Department Head (`trigger.current.opened_by.department.dept_head`), update RITM work notes, and alert IT Procurement Lead.

Step 9: Conditional Decision Branching on Approval Outcome

- **Action:** Flow Logic -> If / Else If
- **Condition Evaluated:** Flow -> Step 7 -> Approval State
- **Branch A:** `Approval State == 'Approved'` -> Proceed to Step 10.
- **Branch B:** `Approval State == 'Rejected'` -> Divert to Step 17B.
- **Branch C:** `Approval State == 'Cancelled'` -> Abort flow execution and archive context.

Step 10: [Branch A - Approved] Update RITM Post-Approval

- **Action:** Update Record on `sc_req_item`.
- **Field Modifications:**
 - `stage = fulfillment`
 - `state = 2` (Work in Progress)
 - `approval = approved`
 - `work_notes = Manager approval confirmed by {Approver.name}. Generating hardware fulfillment and staging tasks.`

Step 11: [Branch A - Approved] Dispatch Approval Confirmation Notification

- **Action:** Send Email
- **Recipient:** `trigger.current.requested_for.email`
- **Subject:** Approved: Your Standard Laptop Request `{trigger.current.number}` has been Authorized
- **Payload:** Informs employee of successful sign-off, confirms order transmission to the Hardware Fulfillment Lab, and provides estimated delivery window.

Step 12: [Branch A - Approved] Automated Task 1 Provisioning — Hardware Staging & Imaging

- **Action:** Create Catalog Task on table `sc_task`.
- **Field Mappings:**
 - `request_item = trigger.current.sys_id`

- `assignment_group = Hardware Fulfillment Group (sys_id: d8e9f0a1b2c34567890abcdef1234567)`
- `short_description = 'Stage, Image, and Asset-Tag Laptop: ' + trigger.current.variables.laptop_model`
- `priority = 3` (Moderate)
- `state = 1` (Open)
- `description` = Concatenated instruction string containing Model, OS, RAM/Storage Bundle, Accessories, and special configuration instructions.
- **Form Variable Formatting:** Exposes catalog variables (`laptop_model`, `operating_system`, `department`, `shipping_address`) directly in the fulfiller task view.
- **Notification:** Triggers Task Assignment Alert (NOTIF-04) to `Hardware Fulfillment Group`.

Step 13: [Branch A - Approved] Wait for Condition — Task 1 Completion & Validation

- **Action:** `Wait for Condition` on `sc_task` (Task 1).
- **Resume Condition:** `sc_task.state IN ('3', '4')` (Closed Complete OR Closed Incomplete).
- **Outcome Verification Logic:**
- *IF `state == 4` (Closed Incomplete):* Intercept closure code. If `Out of Stock`, halt main flow and trigger Subflow: `Procurement Vendor Backorder Purchasing`.
- *IF `state == 3` (Closed Complete):* Validate that `sc_task.cmdb_ci` and `sc_task.asset` are populated; proceed to Step 14.

Step 14: [Branch A - Approved] Automated Task 2 Provisioning — Logistics & Delivery Dispatch

- **Action:** `Update Record` on `sc_req_item`:
- `stage = delivery`
- `work_notes = Hardware staging and imaging completed. Initiating logistics delivery dispatch.`
- **Action:** `Create Catalog Task` on table `sc_task`.
- **Field Mappings:**
- `request_item = trigger.current.sys_id`
- `assignment_group = IT Logistics & Deskside Support (sys_id: e9f0a1b2c3d45678901abcdef2345678)`
- `short_description = 'Package, Ship, and Deliver Laptop to: ' + trigger.current.requested_for.name`
- `priority = 3` (Moderate)
- `state = 1` (Open)
- `description` = Delivery details: Delivery Type, Destination Shipping Address, Contact Phone, and Staged Asset Tag/Serial Number.

Step 15: [Branch A - Approved] Wait for Condition — Task 2 Completion

- **Action:** `Wait for Condition` on `sc_task` (Task 2).
- **Resume Condition:** `sc_task.state == '3'` (Closed Complete).
- **Verification:** Requires technician to input carrier tracking number (or deskside handover signature timestamp) into `sc_task.work_notes` or custom closure field.

Step 16: [Branch A - Approved] Automated Hardware Asset Synchronization

- **Action:** `Update Record` on table `alm_hardware`.
- **Query Match:** `serial_number == Task1.serial_number` OR `asset_tag == Task1.asset_tag`.
- **Field Modifications:**
- `install_status = 1` (In Use)

- `substatus = ''` (Cleared from 'Reserved')
- `assigned_to = trigger.current.requested_for.sys_id`
- `department = trigger.current.requested_for.department`
- `location = trigger.current.requested_for.location`
- **CMDB Relationship:** Creates / confirms relationship `cmdb_rel_ci` linking the Computer CI to the Assigned User (`Runs on::Runs`).
- **RITM Association:** Sets `sc_req_item.configuration_item = alm_hardware.ci`.

Step 17: Lifecycle Closure & Branch Handling

Sub-Step 17A: Approved Happy Path Finalization

1. Update `sc_req_item`:

- `stage = complete`
- `state = 3` (Closed Complete)
- `active = false`
- `work_notes = All fulfillment tasks completed successfully. Asset assigned and device verified delivered.`

1. Evaluate Parent Request (`sc_request`):

- Flow executes GlideAggregate query counting active child RITMs: `request == trigger.current.request AND active == true`.
- *If active child count == 0:* Update `sc_request`:
- `request_state = closed_complete`
- `stage = closed`
- `state = 3` (Closed Complete)
- `active = false`

1. Dispatch Delivery & Sign-off Notification (**NOTIF-05**):

- Sends rich HTML email to employee containing tracking number, IT Service Desk welcome guide, first-time login setup instructions, and hardware sign-off survey link.

Sub-Step 17B: Rejected Exception Path

1. Update `sc_req_item`:

- `stage = request_cancelled`
- `state = 4` (Closed Incomplete)
- `approval = rejected`
- `active = false`

```
work_notes = 'Request rejected by ' + Approver.name + '. Rejection Comments: ' +
Approval.comments
```

1. Update Parent Request (`sc_request`):

- Update `sc_request.request_state = 'closed_incomplete', active = false`.

1. Dispatch Rejection Advisory Notification (**NOTIF-03**):

- Sends formal rejection advisory to employee detailing exact manager comments and instructions on how to resubmit an order with revised business justification.

1. Terminate Flow Execution:

- Flow context transitions to `Completed (With Rejection)`.

4. Approval Logic, Escalation Schedules & Governance Rules

The approval architecture balances executive accountability with operational velocity, preventing request stagnation through strict escalation rules.

```
stateDiagram-v2
    [*] --> EvaluatingApprover: Flow Step 4
    EvaluatingApprover --> StandardManager: Manager Present & Active
    EvaluatingApprover --> GovernanceQueue: Manager Null or Inactive

    StandardManager --> PendingApproval: Generate sysapproval_approver
    GovernanceQueue --> PendingApproval: Assigned to IT Procurement Group

    state PendingApproval {
        [*] --> Requested: T+0h
        Requested --> ReminderFired: T+48h Notification
        ReminderFired --> Escalated: T+120h (5 Days)
        Escalated --> AssignedToDeptHead: Reassign to cmn_department.dept_head
    }

    PendingApproval --> Approved: Line Manager or Dept Head Signs Off
    PendingApproval --> Rejected: Approver Rejects with Comments

    Approved --> [*]: Advance to Fulfillment
    Rejected --> [*]: Abort to Request Cancelled
```

4.1 Separation of Duties & Anti-Tamper Enforcement

- **Self-Approval Prohibition:** If the submitting user (`opened_by`) is identical to the resolved approver (`opened_by.manager`), the flow detects a separation-of-duties conflict. The flow automatically bypasses self-approval and escalates to the second-level manager (`opened_by.manager.manager`).
- **Immutability of Form Data:** Once the approval record is created, a client-side UI Policy locks all catalog item variables (`laptop_model`, `operating_system`, `shipping_address`) as **Read-Only = True**, ensuring specifications cannot be altered while awaiting sign-off.

4.2 Timers and Escalation Matrix

Milestone	Elapsed Time	Action Taken	Recipient / Target	System Audit Annotation
Initial Request	T + 0 Hours	Approval record created in <code>sysapproval_approver</code> .	Line Manager	<code>Approval requested from {Manager.name}</code>
First Reminder	T + 48 Business Hours	Automated notification reminder dispatched.	Line Manager	<code>48-hour reminder sent to {Manager.name}</code>

Executive Escalation	T + 120 Business Hours (5 Days)	Approval record reassigned. Original approval marked Cancelled (Auto-Escalated).	Department Head (cmn_department.dept_head)	Escalated to Dept Head {DeptHead.name} due to SLA inactivity threshold.
Final Expiration	T + 168 Business Hours (7 Days)	Auto-reject request if no sign-off.	Requester & Manager	Request auto-cancelled due to lack of authorization within 7 business days.

5. Automated Catalog Task Provisioning & SLA Schedules

The fulfillment lifecycle decomposes into two distinct operational catalog tasks (`sc_task`), isolating workshop staging from physical distribution.

```
gantt
    title Standard Laptop Order Task Fulfillment SLA Schedule (72 Hours Total)
    dateFormat YYYY-MM-DD HH:mm
    axisFormat %H:%M
    section Task 1: Staging (48h)
    Queue Pickup & Assignment           :a1, 2026-10-01 08:00, 4h
    Imaging, OS Prep & Asset Linking      :a2, after a1, 20h
    Staging QA & Buffer Window           :a3, after a2, 24h
    section Task 2: Logistics (24h)
    Packaging & Courier Handover          :b1, after a3, 8h
    Transit & Deskside Receipt Handover :b2, after b1, 16h
```

5.1 Task 1: Hardware Staging & Configuration Specification

Task Parameter	Configuration Specification
Table	<code>sc_task</code>
Assignment Group	Hardware Fulfillment Group (sys_id: <code>d8e9f0a1b2c34567890abcdef1234567</code>)
Short Description	Stage, Image, and Asset-Tag Laptop: [<code>laptop_model</code>]
Priority	3 - Moderate
Variables Exposed	<code>laptop_model</code> , <code>operating_system</code> , <code>ram_storage_bundle</code> , <code>accessories</code> , <code>department</code> , <code>requested_for</code>
Mandatory Exit Conditions	Fulfiller must link valid <code>asset</code> (<code>alm_hardware</code>) and <code>serial_number</code> . Task cannot be marked <code>Closed Complete</code> with an empty asset field.
Task SLA Schedule	48 Business Hours (Schedule: <code>8x5 Mon-Fri excluding Holidays</code>).

5.2 Task 2: Logistics & Deployment Delivery Specification

Task Parameter	Configuration Specification
Table	sc_task
Assignment Group	IT Logistics & Deskside Support (sys_id: e9f0alb2c3d45678901abcdef2345678)
Short Description	Package, Ship, and Deliver Laptop to: [requested_for.name]
Priority	3 - Moderate
Variables Exposed	delivery_type, shipping_address, contact_phone, requested_for, asset_tag
Mandatory Exit Conditions	Fulfiller must enter shipping tracking number (e.g., FedEx/UPS) or physical handover confirmation in task work notes.
Task SLA Schedule	24 Business Hours (Schedule: 8x5 Mon-Fri excluding Holidays).

6. Automated Notification Architecture

The solution deploys five standardized, responsive HTML email notifications triggered at specific milestone boundaries.

```
classDiagram
    class NotificationEngine {
        +NOTIF-01: Submission Receipt
        +NOTIF-02: Manager Approval Request
        +NOTIF-03: Rejection Advisory Notice
        +NOTIF-04: Task Assignment Alert
        +NOTIF-05: Delivery Dispatch Notice
    }
    class Recipient {
        +Requester (requested_for)
        +Line Manager (opened_by.manager)
        +Fulfillment Groups (sys_user_group)
    }
    NotificationEngine --> Recipient : Dispatches to
```

6.1 Master Notification Catalog

ID	Notification Name	Trigger Milestone	Recipient (To:)	Subject Line Specification	Key Content & Actionable Tokens
----	-------------------	-------------------	-----------------	----------------------------	---------------------------------

||
||
||

NOTIF-01	Submission Receipt	Step 3: Flow Trigger committed	<code>requested_for.email</code>	Order Confirmation: Standard Laptop Request - <code>\${number}</code>	RITM number, selected model, OS, delivery location, 72-hour fulfillment commitment, Service Portal tracking link.
NOTIF-02	Manager Approval Request	Step 6: Approval record generated	<code>approver.email</code>	ACTION REQUIRED: Laptop Approval Request for <code>\${requested_for.name}</code> - <code>\${number}</code>	Detailed hardware specs, cost center code, business justification, interactive one-click "Approve" / "Reject" buttons.
NOTIF-03	Rejection Advisory Notice	Step 17B: Manager rejects	<code>requested_for.email</code>	Notice: Standard Laptop Request <code>\${number}</code> has been Declined	Disapproval decision, verbatim manager comments, instructions for appeal or re-submitting with modified justification.
NOTIF-04	Task Assignment Alert	Steps 12 & 14: Task created	<code>assignment_group.email</code>	New Fulfillment Task: <code>[Task Short Description]</code> - <code>\${number}</code>	Direct link to SCTASK record, exposed catalog variables,
					priority,
					customer due date SLA.
NOTIF-05	Delivery Dispatch Notice	Step 17A: RITM closure	<code>requested_for.email</code>	Delivered: Your Standard Laptop is Ready - Welcome!	Courier tracking URL, physical handover confirmation, IT Welcome Pack link, day-one login checklist, IT Support contact info.

6.2 Actionable Email Markup Template (Approval Request Example)

```

<div style="font-family: Arial, sans-serif; max-width: 600px; margin: 0 auto; border: 1px solid #e0e0e0; border-radius: 8px; overflow: hidden;">
  <div style="background-color: #032d42; color: #ffffff; padding: 20px; text-align: center;">
    <h2 style="margin: 0;">IT Procurement Authorization Request</h2>
  </div>
  <div style="padding: 24px; color: #333333; line-height: 1.6;">
    <p>Hello <strong>${approver.first_name}</strong>,</p>
    <p>A standard laptop procurement request has been submitted for an employee in your reporting hierarchy and requires your authorization.</p>

    <table style="width: 100%; border-collapse: collapse; margin: 20px 0;">
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Request Number:</td><td>${sysapproval.number}</td></tr>
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Requested For:</td><td>${sysapproval.requested_for.name}</td></tr>
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Department:</td><td>${sysapproval.requested_for.department.name}</td></tr>
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Laptop Model:</td><td>${sysapproval.variables.laptop_model}</td></tr>
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Operating System:</td><td>${sysapproval.variables.operating_system}</td></tr>
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Cost Center:</td><td>${sysapproval.requested_for.cost_center.code}</td></tr>
      <tr style="border-bottom: 1px solid #eeeeee;"><td style="padding: 8px 0; font-weight: bold;">Justification:</td><td>${sysapproval.variables.business_justification}</td></tr>
    </table>

    <div style="text-align: center; margin: 30px 0;">
      <a href="mailto:${instance_email}?subject=Re: ${sysapproval.number}%20approve&body=approve"
        style="background-color: #2e7d32; color: #ffffff; padding: 12px 24px; text-decoration: none; border-radius: 4px; font-weight: bold; margin-right: 15px;">APPROVE</a>
      <a href="mailto:${instance_email}?subject=Re: ${sysapproval.number}%20reject&body=reject"
        style="background-color: #c62828; color: #ffffff; padding: 12px 24px; text-decoration: none; border-radius: 4px; font-weight: bold;">REJECT</a>
    </div>

    <p style="font-size: 12px; color: #777777; text-align: center;">You may also review and action this request directly in the <a href="${sysapproval.URI_REF}">ServiceNow Employee Center</a>.</p>
  </div>
</div>

```

7. Native Error Handling & Fault-Tolerance Framework

To satisfy enterprise resilience requirements, the workflow embeds a dedicated **Flow Designer Error Handler** block that intercepts any unexpected runtime failures.

```

flowchart TD
    subgraph Flow_Execution_Boundary [Flow Designer Main Execution Scope]
        StepA[Action Steps 1 to 17] -->|Runtime Exception / DB Timeout| ThrowErr[Exception Thrown]
    end
    end

```

```

subgraph Error_Handler_Block [Native Error Handler Block]
  ThrowErr --> Catch[Catch Block Activated]
  Catch --> E1[Action 1: Write Forensic Audit Entry to syslog]
  Catch --> E2[Action 2: Update sc_req_item Work Notes]
  Catch --> E3[Action 3: Generate Priority 2 Incident Ticket]
  Catch --> E4[Action 4: Dispatch Emergency Alert to Platform Admins]
end

E3 --> Remediate[Platform Support Engineering Triages Incident]
Remediate --> Resume[Resume Flow Execution from Last Saved Context]

```

7.1 Catch Block Actions & Failure Recovery

Error Action Step	Invoked Action	Parameters & Operational Logic	Expected System Outcome
Action E1	Log (Syslog)	Log Level: Error Message: Flow 'Standard Laptop Procurement' failed on Step {step_number}. Context ID: {sys_flow_context.sys_id}. Exception: {error_message}	Writes permanent error trace to syslog for engineering diagnosis.
Action E2	Update Record (sc_req_item)	Field: work_notes Value: SYSTEM EXCEPTION: Automated workflow halted unexpectedly. Platform Support Incident generated. Requester action not required.	Provides immediate transparency on the RITM form without exposing raw stack traces to end users.
Action E3	Create Record (incident)	caller_id = System Administrator assignment_group = ServiceNow Platform Support priority = 2 - High category = Enterprise Software short_description = Flow Failure: Standard Laptop Procurement on {trigger.current.number} description = Concatenated dump of Flow Context ID, failed action name, input payload, and error string.	Ensures formal ITIL incident management response with associated MTTR SLAs.
Action E4	Send Email	To: sn_platform_admins@enterprise.com Subject: CRITICAL ALERT: Flow Execution Aborted on {trigger.current.number}	Dispatches immediate alert to duty engineers for high-visibility escalations.

7.2 Database Transaction Integrity & Checkpoint Resumption

Because Flow Designer executes each action within managed database transactional scopes, any unexpected node restart or database lock timeout records the last uncommitted checkpoint in **sys_flow_context**. Once the underlying issue is resolved by platform support, the engineer can execute the native **Test & Resume** action directly from the Flow Execution Context viewer, replaying the failed action without duplicating previously completed catalog tasks or re-requesting manager approval.

8. Document Sign-off & Revision History

Version	Date	Author	Role	Description of Change
1.0	2026-09-30	worker_phase3	Phase 3 Design Lead	Authoritative Deliverable: 17-Step End-to-End Workflow Design, Trigger Mechanics, Escalations, Task Definitions, Notification Catalog, and Error Handling.

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